

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

RAM Reseating – Step-by-Step Documentation Steps performed:

1. Shut down the desktop/laptop completely and disconnected the power cable.
2. Opened the computer case/back cover and located the **RAM slots on the motherboard.**
3. Pressed the locking clips on both sides of the RAM slot to release the RAM module.
4. Carefully removed the RAM module by holding it from the edges.
5. Checked the RAM and slot for **dust, damage, or loose connections.**
6. Aligned the notch on the RAM with the slot and inserted it firmly.
7. Pressed down until the side clips locked the RAM into place.
8. Closed the case/back cover, reconnected the power, and started the computer.
9. Checked whether the computer booted normally and whether the RAM was detected.

Issues observed:

No physical damage was observed. The RAM was slightly loose before reseating. After reseating, the computer started normally and the RAM was detected.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Step	Troubleshooting Action	What to Check
1	Identify the symptom	Check for no display, screen artifacts, flickering, crashes, or low gaming performance.
2	Check display connections	Make sure the monitor cable is firmly connected to the GPU and monitor. Try another cable or port if available.
3	Check GPU physical connection	Power off the PC and ensure the GPU is properly seated in the PCIe slot. Check that GPU power connectors are firmly connected.
4	Check GPU drivers	Open Device Manager and check for GPU driver errors. Update or reinstall the graphics driver if necessary.
5	Check GPU temperature	Monitor GPU temperature during use. Excessive temperature may indicate poor cooling, blocked fans, or dust.
6	Test the GPU	Run a graphics/game test and check whether display problems, crashes, or artifacts continue.
7	Confirm the fix	Restart the PC and run a game or graphics test. Confirm that the display is stable and the GPU works without errors.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Stage	What was done	Error/BIOS indication
1. Before disconnecting	Checked the storage drive while the PC was working normally.	BIOS detected the SSD/HDD and Windows booted normally.
2. Drive disconnected	Powered off the test PC and disconnected the SATA data/power cable (or removed the NVMe drive).	On startup, BIOS showed “ No Boot Device Found ” / “ Storage Device Not Detected. ”
3. Boot attempt	Tried to continue booting without the drive.	The system could not find the operating system and did not boot into Windows.
4. Drive reconnected	Powered off the PC and securely reconnected/reseated the drive.	BIOS detected the SSD/HDD again.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Reason: Faulty, loose, or incompatible RAM can cause random restarts, crashes, freezing, or failure to boot. Since the problem occurs randomly, RAM is a good first hardware component to test.

Diagnostic Steps for RAM

1. **Shut down the PC** and disconnect the power cable.
2. **Open the computer case** and locate the RAM modules.
3. **Check the RAM physically** for dust, damage, or improper seating.
4. **Reseat the RAM** by removing it and firmly inserting it back into the slot.
5. **Test one RAM module at a time** if multiple modules are installed.
6. **Try another RAM slot** to check whether the slot is causing the problem.
7. **Run Windows Memory Diagnostic** to check for memory errors.
8. **Test the PC under normal use/gaming** and observe whether random restarts continue.